QUANTUM CRYPTOGRAPHY (BB84 PROTOCOL) — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: April 15, 2026

Diagram-based questions on Quantum Cryptography (BB84 Protocol)

Question 1 10.0 points

Examine the Bloch sphere representation of the BB84 protocol quantum states in the diagram below. Describe the position of each of the four states ($|0\rangle$, $|1\rangle$, $|+\rangle$, $|-\rangle$) in terms of their polar and azimuthal angles on the Bloch sphere using rectilinear and diagonal bases.

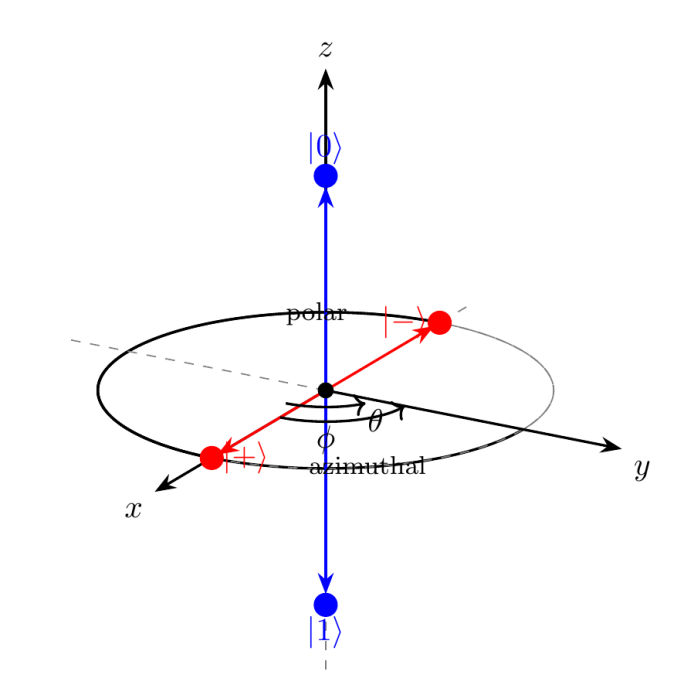

Fig. 1. Bloch sphere representation showing the quantum states used in the BB84 protocol. The sphere should be centered with axes labeled x, y, and z. Place the state |0⟩ at the north pole (z=+1), |1⟩ at the south pole (z=-1), |+⟩ on the equator in the x direction (x=+1), and |-⟩ on the equator in the negative x direction (x=-1). Each state should be labeled with its ket notation. Include polar and azimuthal angle annotations for each state, indicating their positions in both rectilinear and diagonal bases. IMPORTANT: The diagram is missing required labels. The diagram is missing the required polar and. Add each missing label as visible text at the appropriate location in the diagram. IMPORTANT correction needed: The diagram explicitly labels the "rectilinear basis. Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only.

Question 2

10.0 points

Interpret the key exchange sequence in the diagram below for the BB84 protocol between Alice and Bob, including an intercept-resend attack by Eve. Explain how the protocol detects eavesdropping.

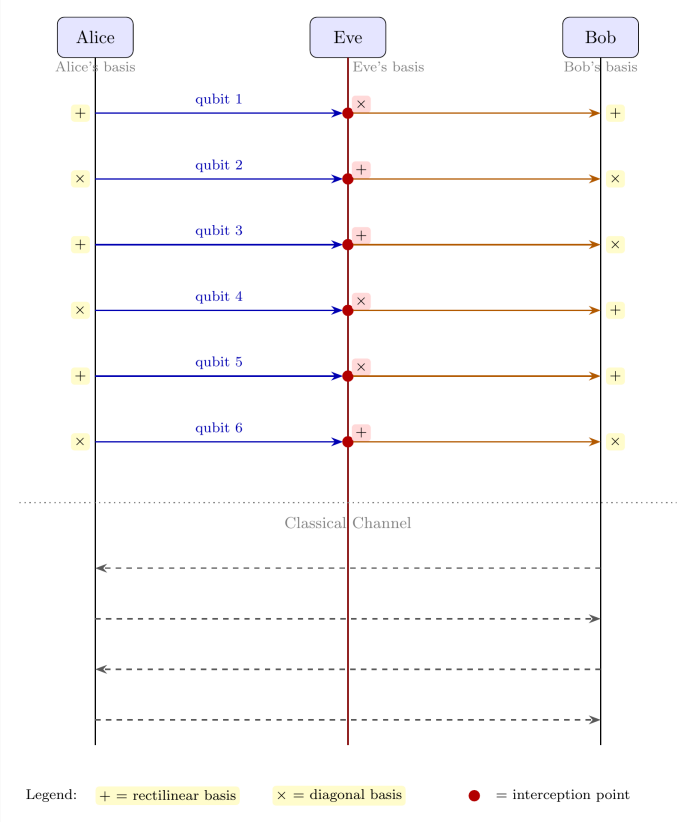

Fig. 2. CUMULATIVE LEAK CORRECTIONS (all must be fixed): The diagram contains a label that explicitly states the conclusion the; The diagram explicitly labels the steps of the classical communication phase. A diagram illustrating the BB84 key exchange protocol sequence between Alice and Bob, including an intercept-resend attack by Eve. The diagram should show: 1. Alice sending a sequence of qubits to Bob, with each qubit randomly prepared in one of two bases (rectilinear or diagonal). 2. Eve intercepting the qubits, measuring them in a randomly chosen basis, and resending them to Bob. 3. Bob receiving the qubits and measuring them in a randomly chosen basis. 4. The classical communication between Alice and Bob to compare bases and detect discrepancies caused by Eve's interference. Include labels for Alice, Bob, Eve, qubit sequence, basis choices, and classical communication. IMPORTANT correction needed: The diagram contains a label that explicitly states the conclusion the. Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only. IMPORTANT correction needed: The diagram explicitly labels the steps of the classical communication phase. Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only.

Question 3

10.0 points

Examine the quantum key distribution channel in the diagram below, which illustrates the probability of eavesdropping detection versus sifted key error rate. Describe the trade-off between security and error rate.

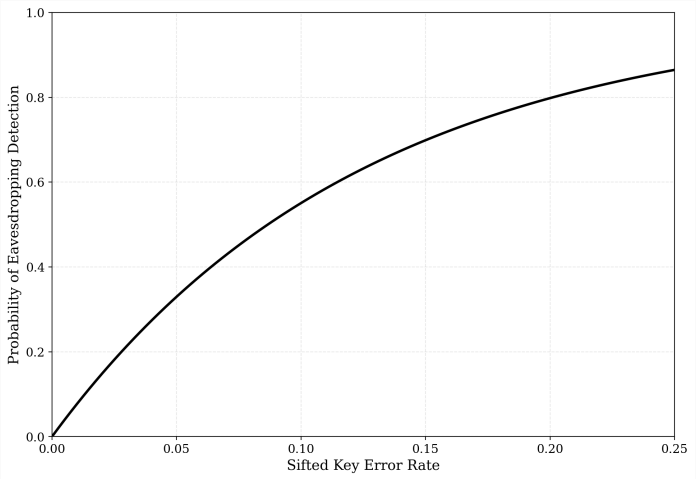

Fig. 3. A plot showing the probability of eavesdropping detection on the y-axis versus the sifted key error rate on the x-axis. The plot should illustrate the trade-off between security (eavesdropping detection probability) and error rate, with a curve that typically shows increasing detection probability with higher error rates. Include labeled axes and a title 'Eavesdropping Detection Probability vs. Sifted Key Error Rate'. IMPORTANT: The diagram is missing required labels. The diagram is missing the title 'Eaves. Add each missing label as visible text at the appropriate location in the diagram. IMPORTANT: The diagram is missing required labels. The diagram is missing the required title as specified in. Add each missing label as visible text at the appropriate location in the diagram.

Question 4

10.0 points

Evaluate the quantum circuit shown below that demonstrates the no-cloning theorem. Explain how this circuit prevents the cloning of an arbitrary quantum state.

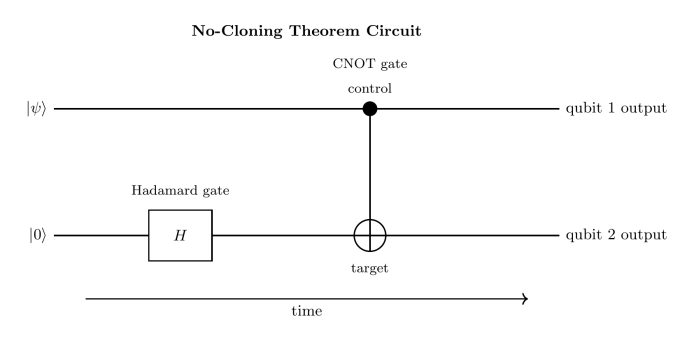

Fig. 4. Quantum circuit demonstrating the no-cloning theorem with two qubit lines labeled |ψ⟩ and |0⟩. The circuit includes a Hadamard gate on the |0⟩ line, followed by a CNOT gate with the control on the |ψ⟩ line and target on the |0⟩ line. The output lines are labeled to show the result of the circuit operation. IMPORTANT correction needed: The diagram contains two separate annotations. Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only. IMPORTANT: Values or sequences in the diagram don't match the question. The diagram shows a Hadamard gate on the. Regenerate using EXACTLY the values and sequences specified.

Question 5

10.0 points

Interpret the correlation curve between sifted key error rate and eavesdropping probability in the quantum key distribution session shown in the diagram below. How can this curve guide secure communication practices?

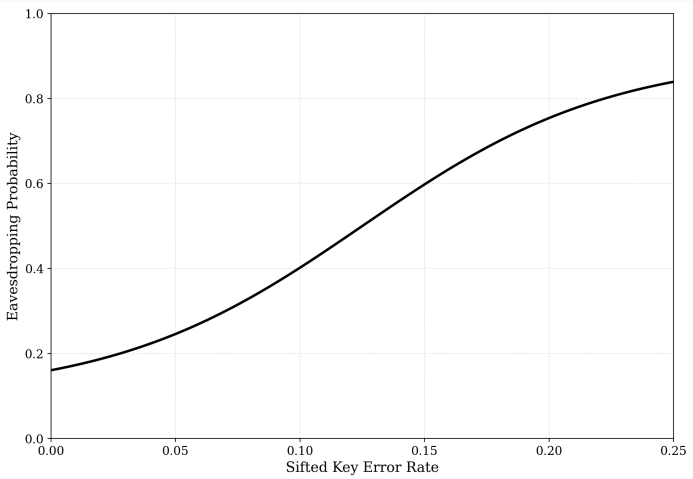

Fig. 5. A correlation curve showing the relationship between sifted key error rate (x-axis) and eavesdropping probability (y-axis) in a quantum key distribution session. The curve should illustrate how an increase in error rate correlates with an increase in eavesdropping probability, highlighting the importance of monitoring error rates for secure communication.